Miniproject 3: Folded Cascode differential Amplifier

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https://github.com/James-Jagielski/MADVLSI/tree/main/Miniproject3

1 Schematics

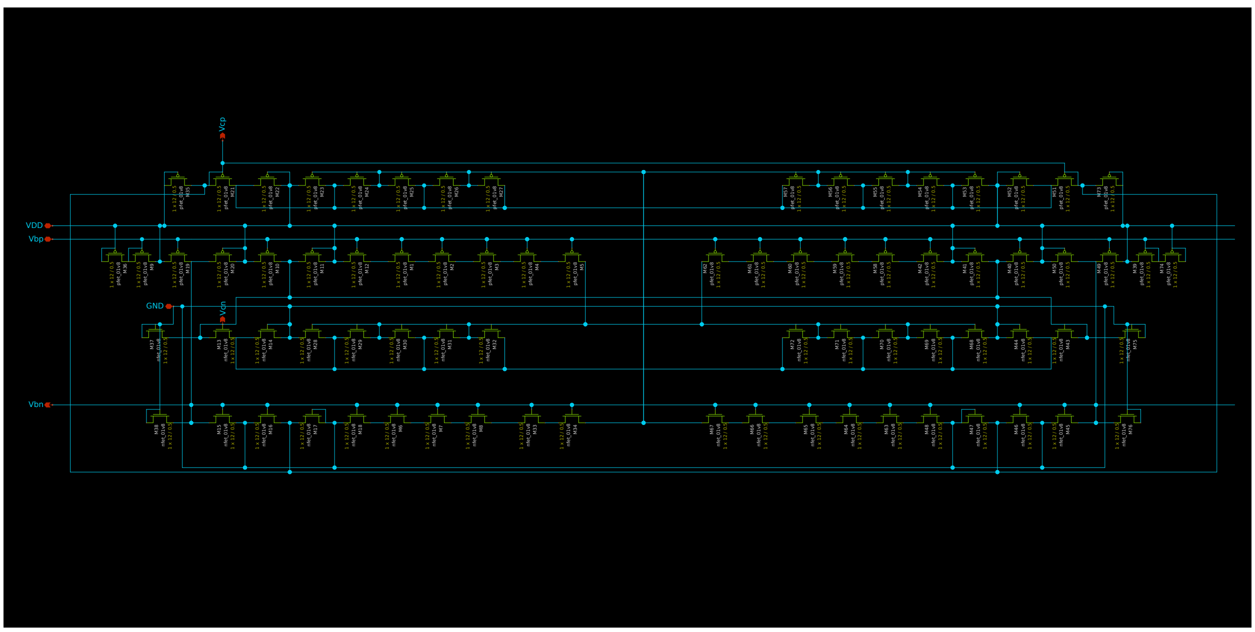


Figure 1: Voltage Bias schematic.

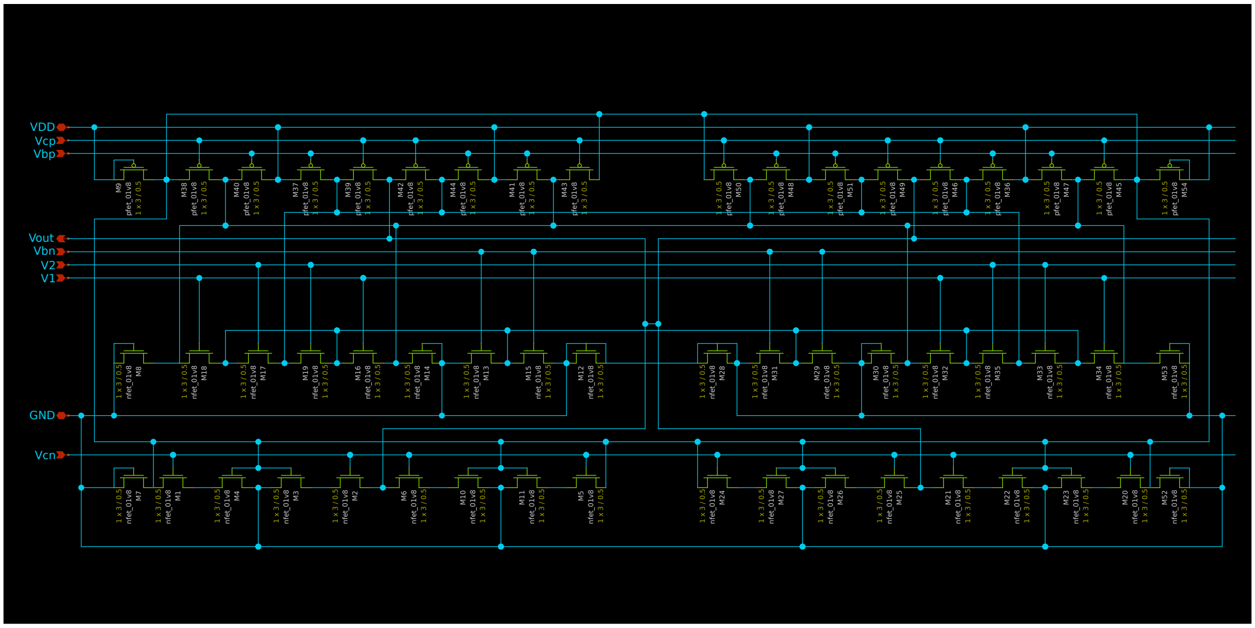


Figure 2: Amplifier Schematic.

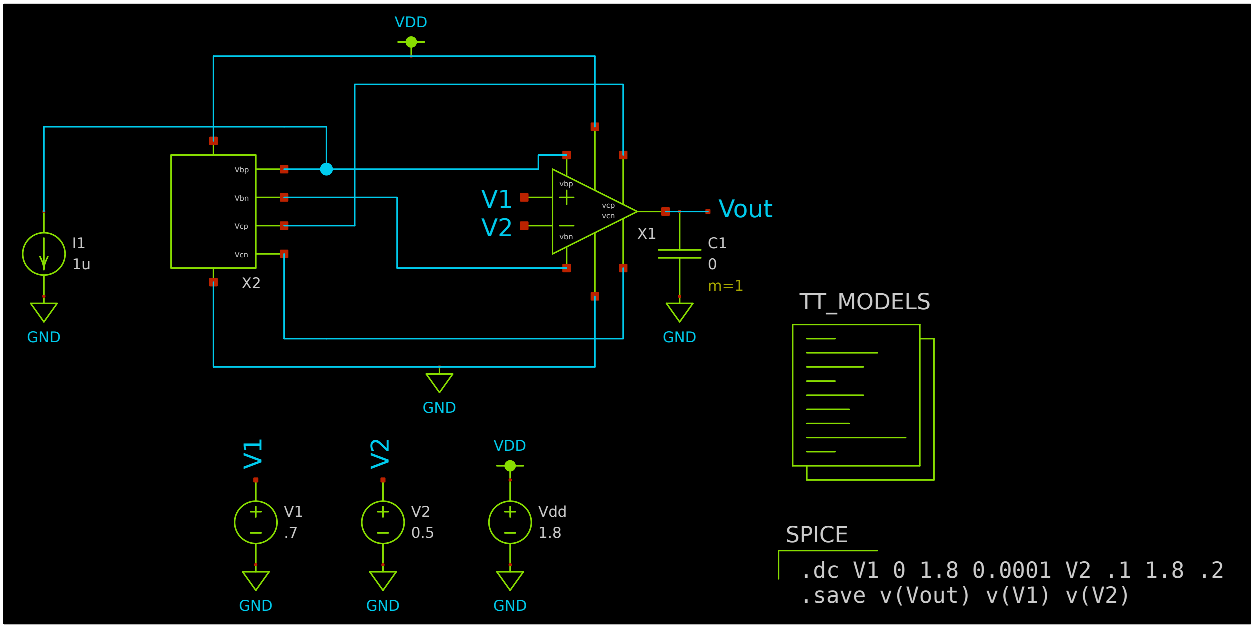


Figure 3: Voltage transfer characteristic schematic.

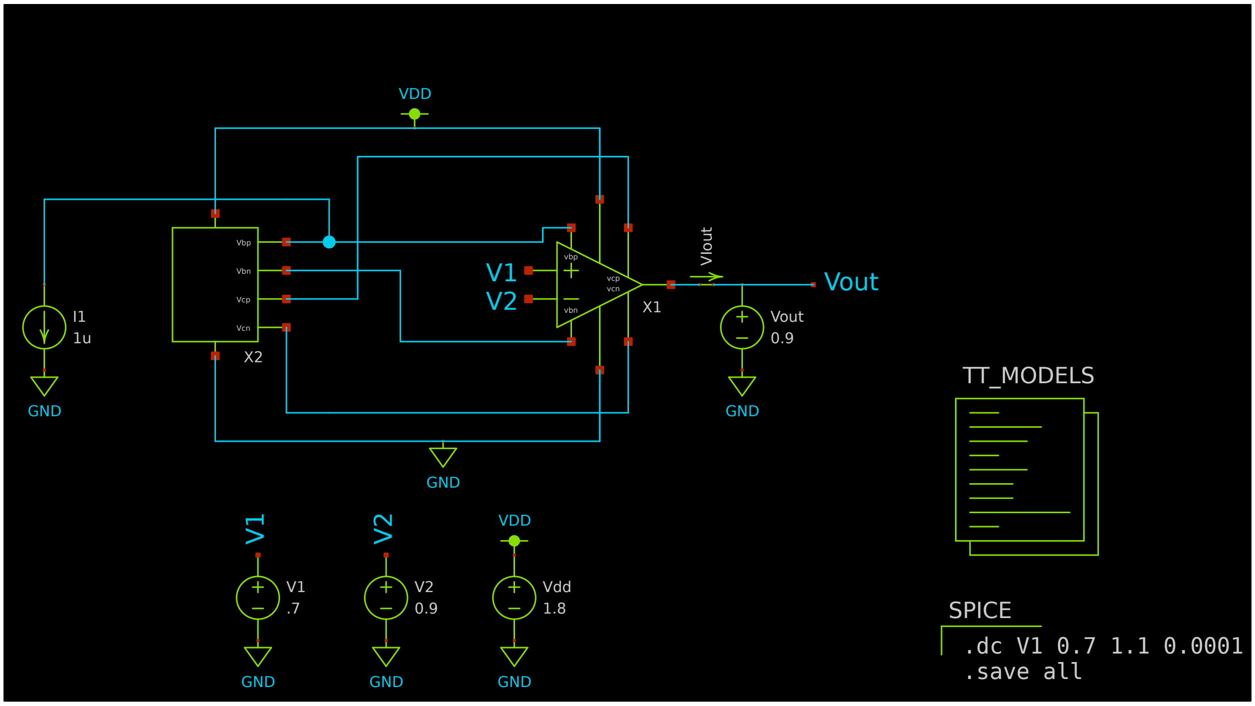


Figure 4: Voltage to current characteristic schematic.

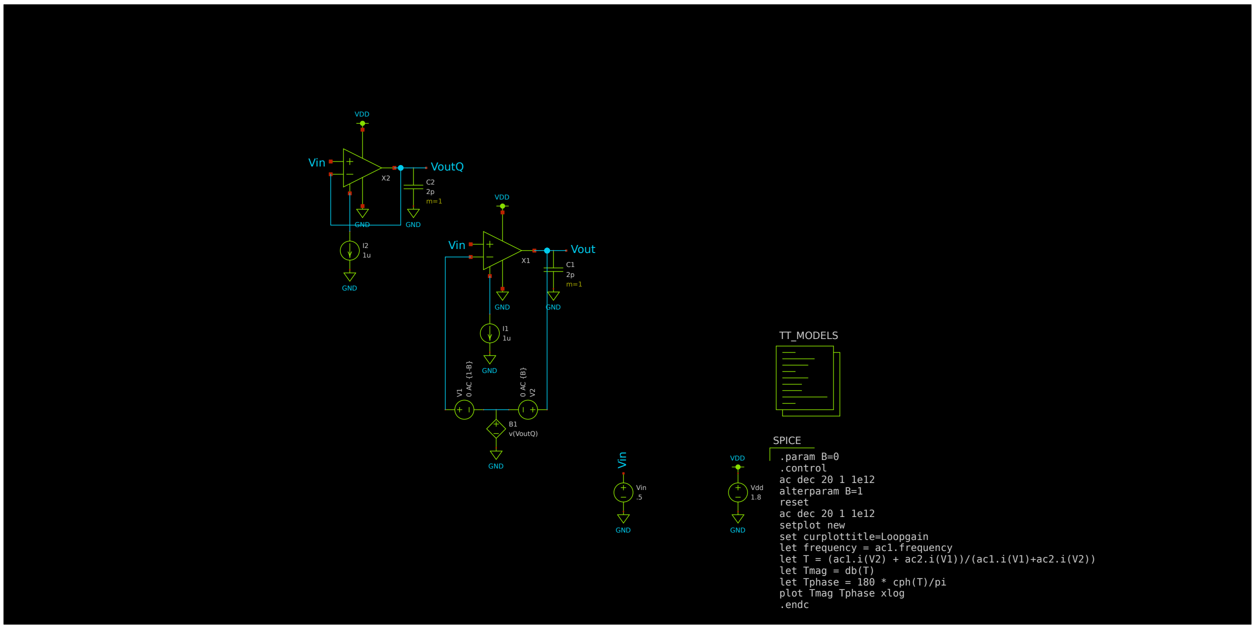


Figure 5: Loop Gain Schematic.

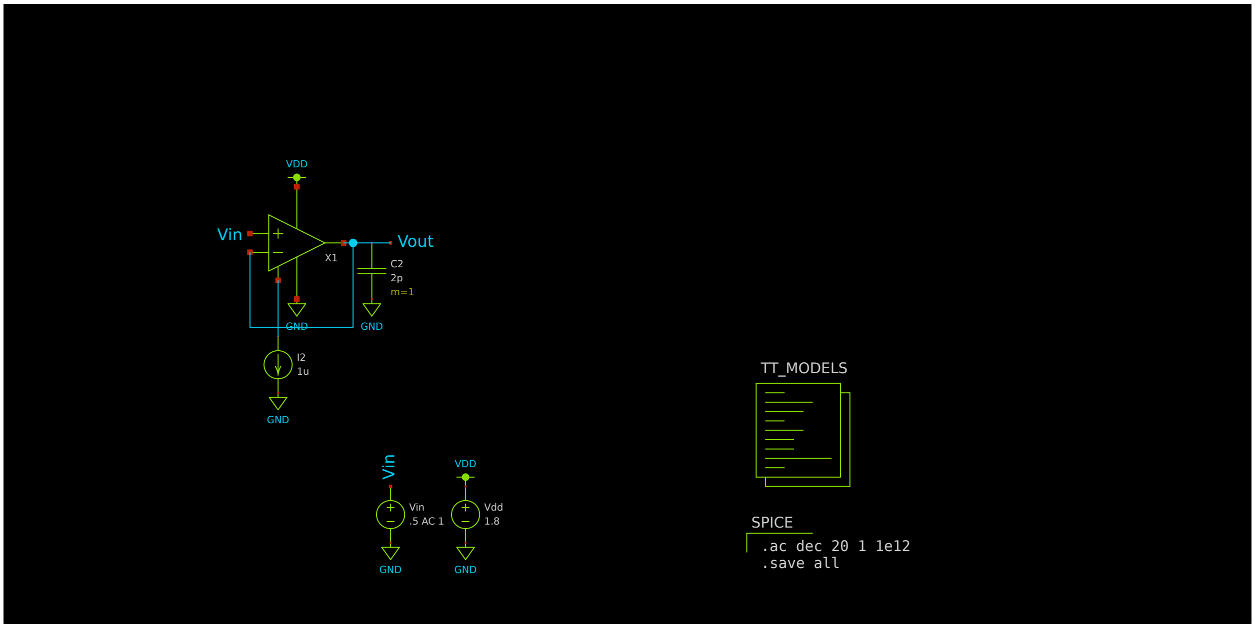


Figure 6: Unity gain follower frequency response schematic.

2 Simulations

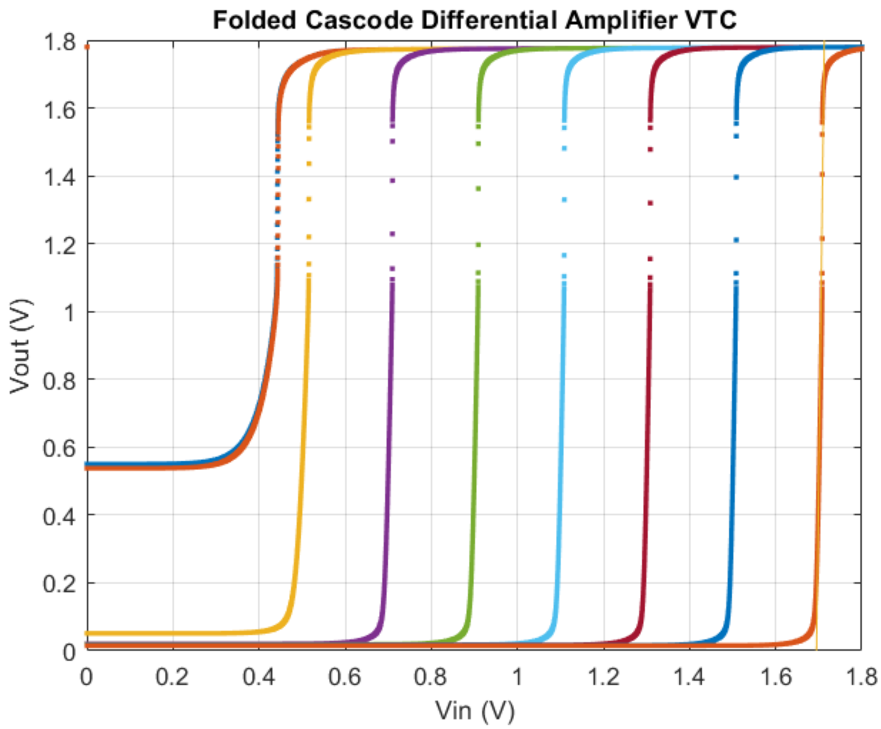


Figure 7: Voltage Transfer Characteristic simulation. I swept the voltage reference V2 from .1V to 1.7V going up by .2V. The curves represent these reference voltages going from left to right.

We can extract the DC gain from the circuit by finding the slope of the VTCs. We will find the slope decreasing as we decrease the voltage on V2. The DC gain is 99.6.

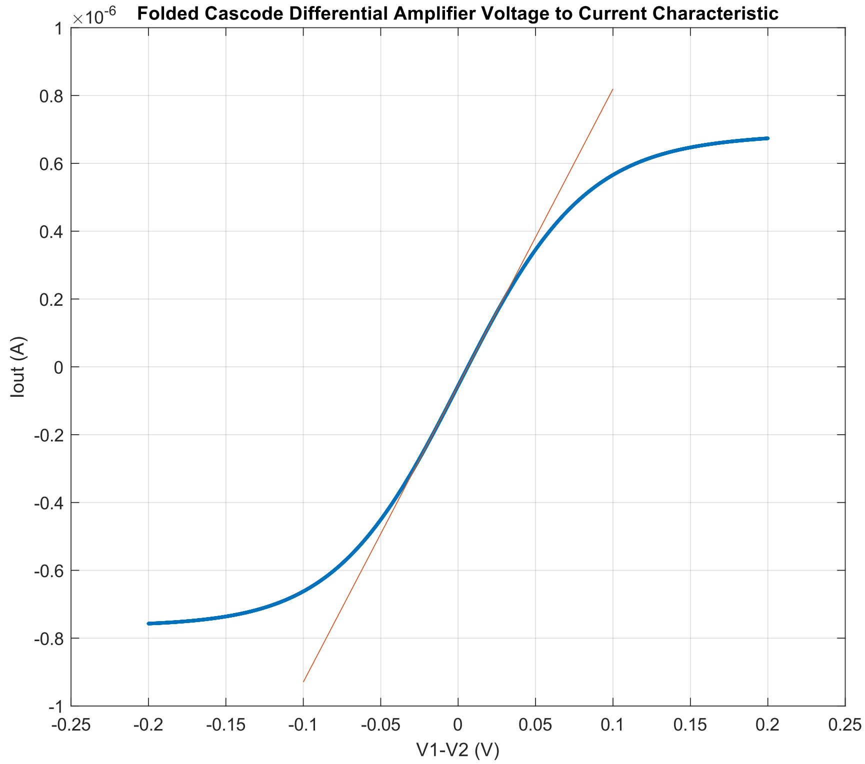


Figure 8: Voltage to Current Characteristic simulation.

The incremental transconductance gain of the circuit is the slope of the voltage to current characteristic, which yields $8.745227 \cdot 10^{-6}$ S or $\frac{A}{V}$. The limiting values of the output current are $-7.570070 \cdot 10^{-7}A$ and $6.738150 \cdot 10^{-7}A$.

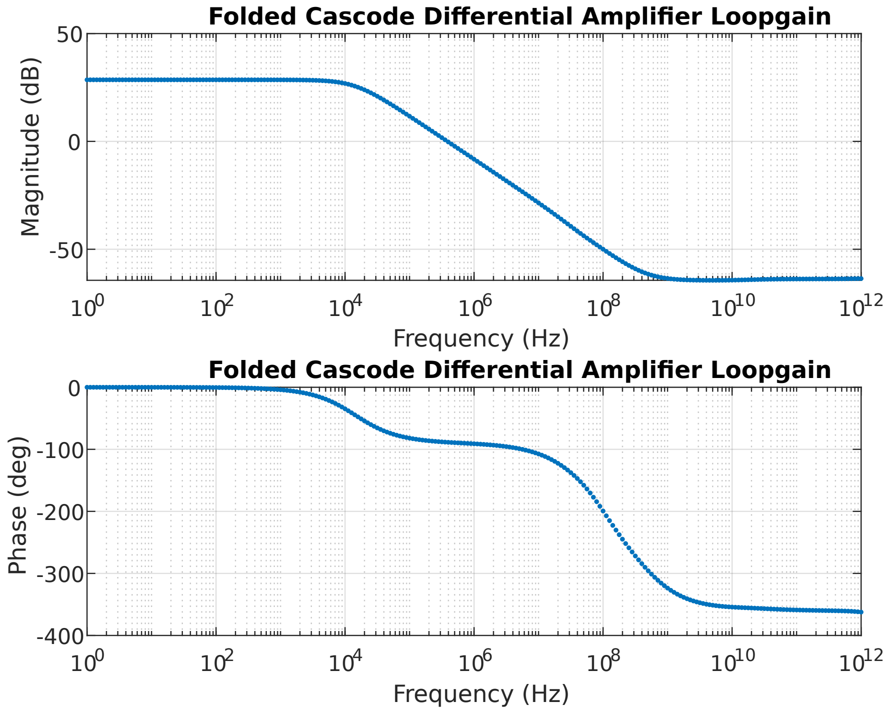


Figure 9: Loopgain simulation.

The low frequency gain can be calculated with the decibal conversion of,

$$dB = 20 \cdot log\frac{Vout}{Vin}. \tag{1}$$

The low frequency gain was 26.776125 (Hz). The unity gain crossover frequency is $4.466836 \cdot 10^5$ (Hz). We would expect $6.9592 \cdot 10^5$ (Hz) from the equation,

$$f_{crossover} = \frac{G_M}{C \cdot 2\pi}. \tag{2}$$

I would expect these values to be much closer, but they are in the same range, so I would say these are pretty comparable results. There is 35% error between the two numbers.

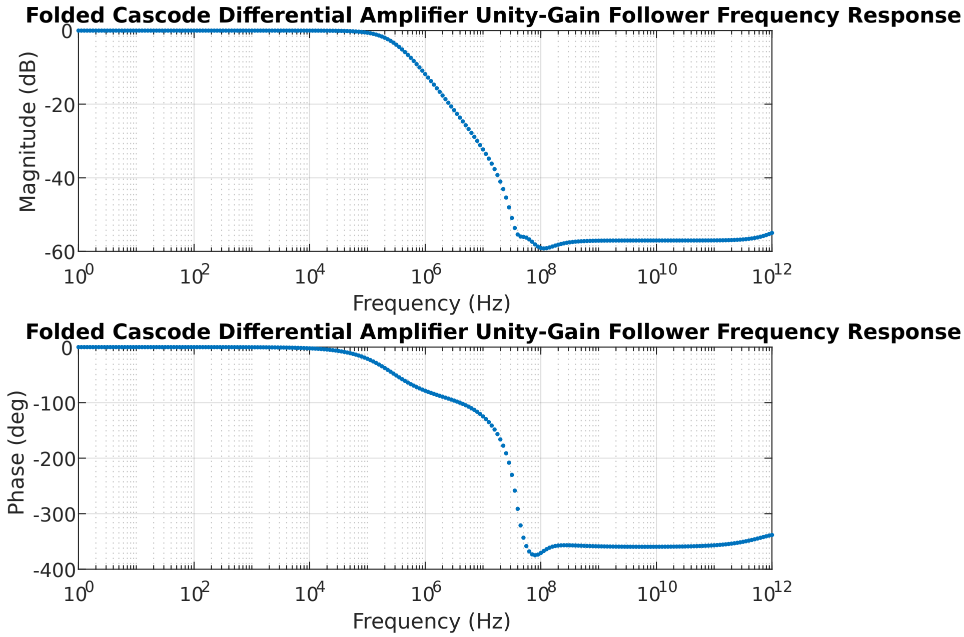


Figure 10: Unity gain follower frequency response simulation.

The corner frequency is $2.818383 \cdot 10^5 Hz$, which is when the gain goes down by -3 dB. This is much less than the cross over frequency of the loopgain simulation. There is 37% error between the the two numbers.

Layout

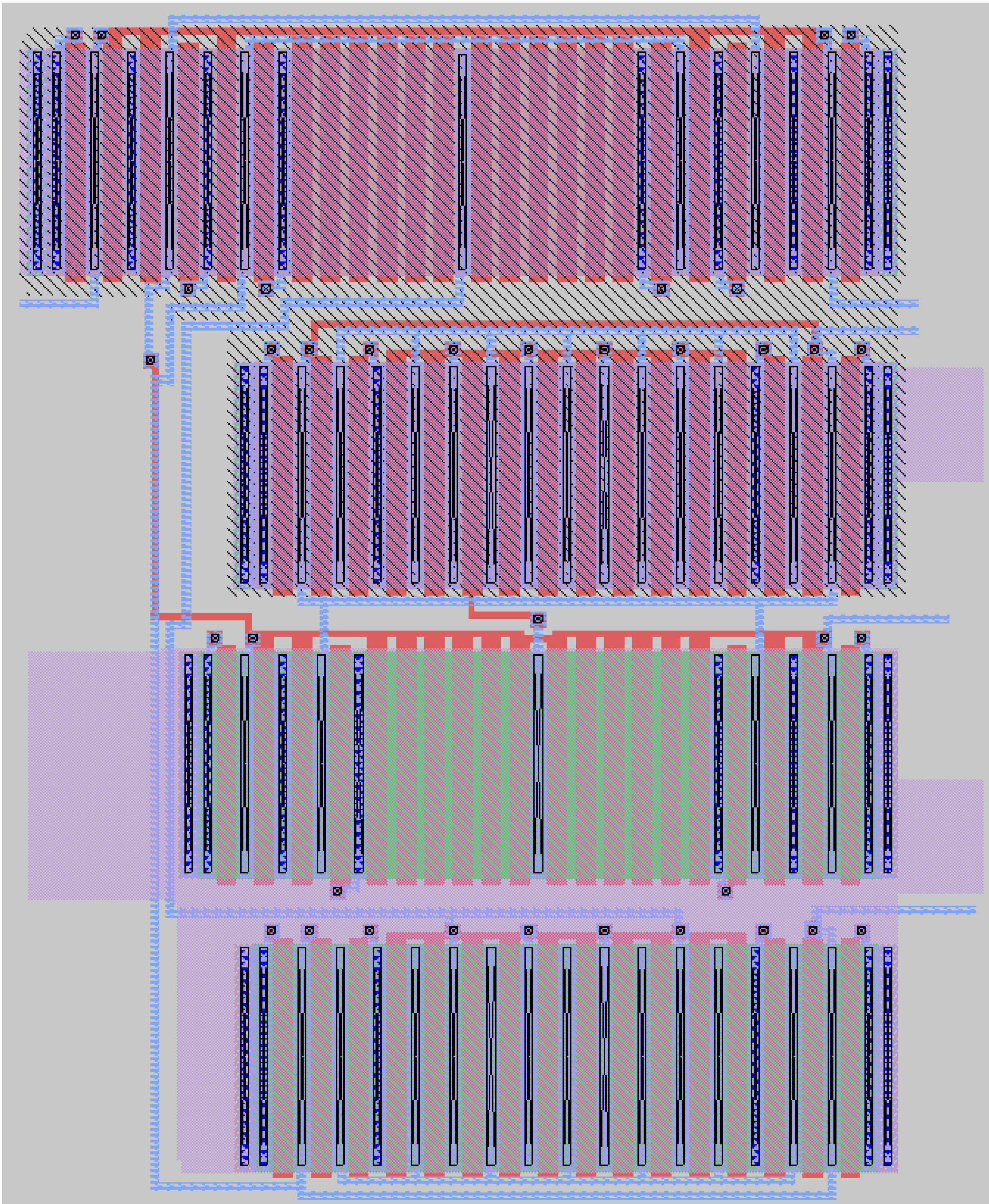


Figure 11: Voltage bias generation layout in MAGIC.

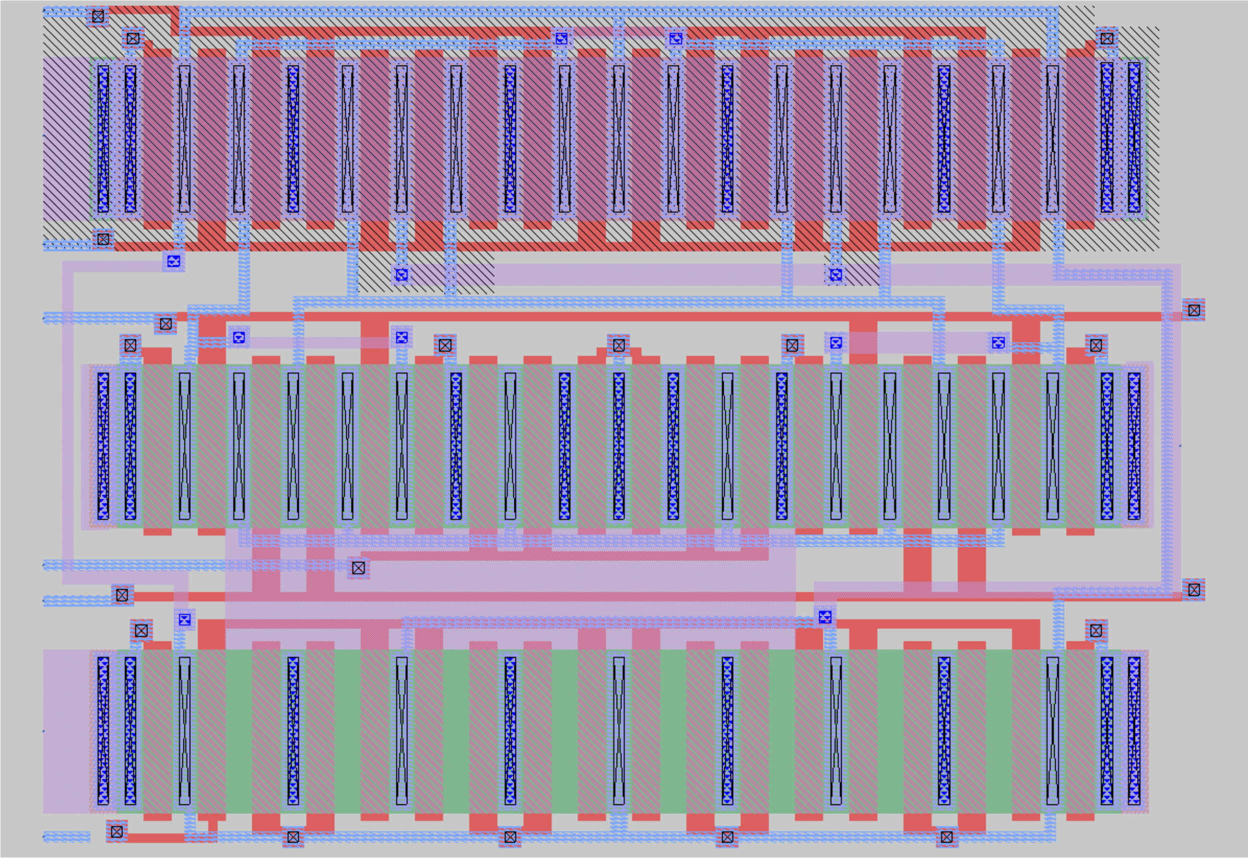


Figure 12: Differential amplifier layout in MAGIC.

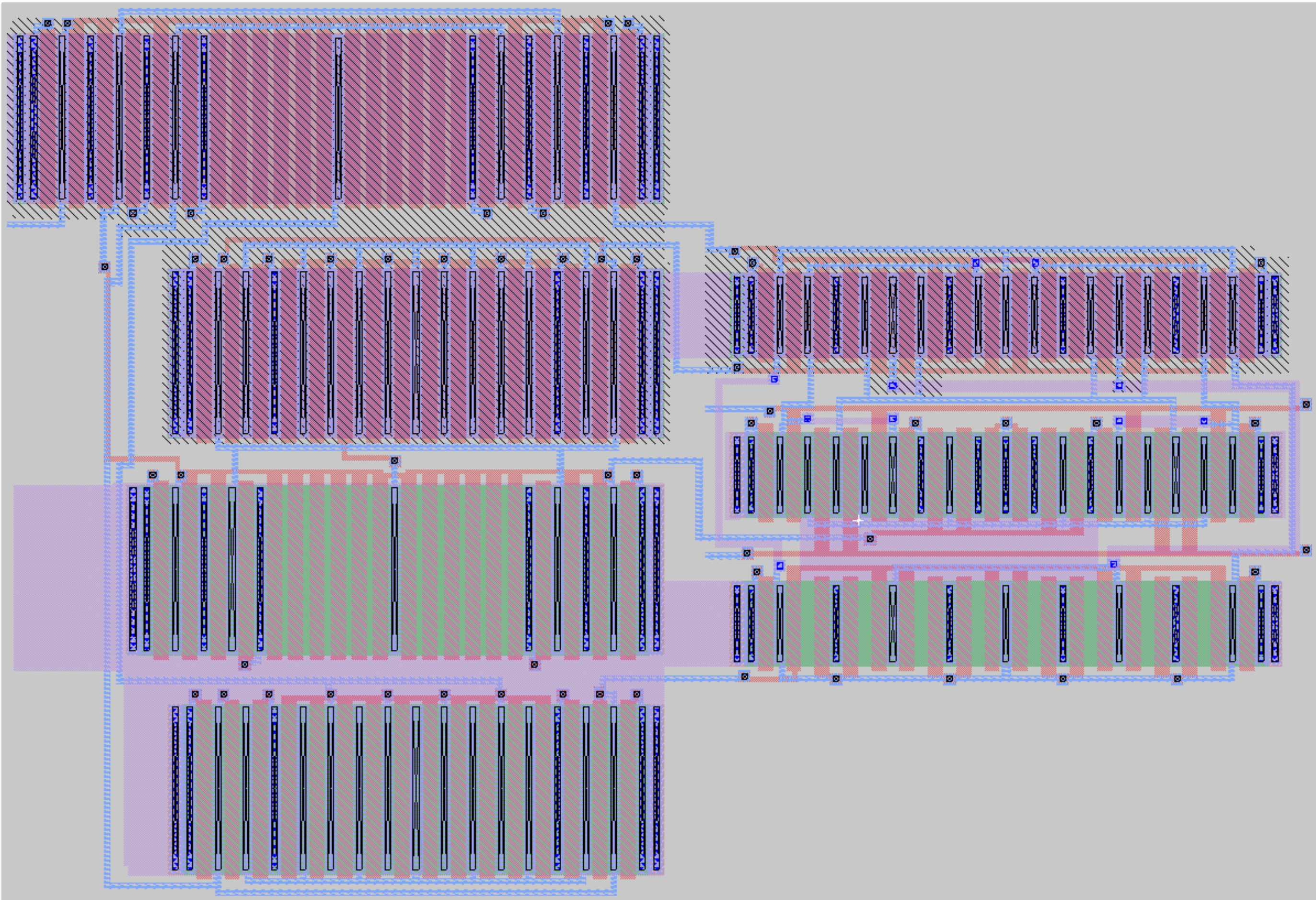


Figure 13: Folded Cascode differential amplifier layout in MAGIC.

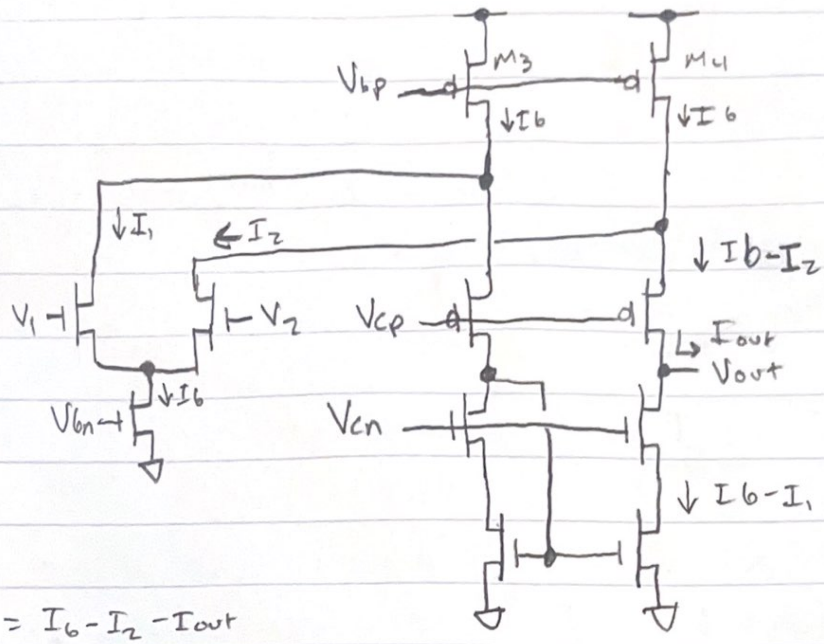
LVS

```
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:44 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M12:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:64 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M1:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:57 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M58:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:4 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M4:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:19 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M61:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:68 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M2:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:40 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M59:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:39 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M60:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:3 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M3:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:31 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M5:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cascode_voltage_bias_0/sky130_fd_pr__pfet_01v8:18 vs. LDS_Cascode_bias_voltage_generator:1/sky130_fd_pr__pfet_01v8:M62:
 w circuit1: 6   circuit2: 12   (delta=66.7%, cutoff=1%)
Cells have no pins;  pin matching not needed.
Device classes Folded_cascode_differential_amplifier.spice and cmdiffamp.spice are equivalent.

Final result: Circuits match uniquely.
Property errors were found.

The following cells had property errors:
 Folded_cascode_differential_amplifier.spice
```

Figure 14: Layout versus schematic output from NETGEN.

madVLSI miniproject 3 Grok The circuit

1.) A)



$I_b - I_1 = I_b - I_2 - I_{out}$

$I_{out} = (\cancel{I_b} - I_2) - (\cancel{I_b} - I_1)$

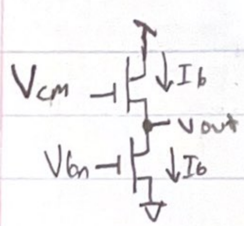
$\boxed{I_{out} = I_1 - I_2}$

when $V_1 > V_2$ Then $I_{out} \approx I_1$. when $V_1 < V_2$ Then $I_{out} \approx -I_2$

Thus, V_1 is The non-inverting input and V_2 is The inverting input.

B) common mode input voltage range.

we can treat The differential pair as a source follower.



$$EKV \Rightarrow I_b = S I_s \log^2\left(1 + e^{\kappa(V_{cm} - V_{T0}) - V_{out}/2U_T}\right)$$

$$\Rightarrow I_b = S I_s \log^2\left(1 + e^{\kappa(V_b - V_{T0})/2U_T}\right)$$

$$\Rightarrow e^{\kappa(V_{cm} - V_{T0}) - V_{out}/2U_T} = e^{\kappa(V_b - V_{T0})/2U_T}$$

$$\Rightarrow \frac{\kappa(V_{in} - V_{T0}) - V_{out}}{\cancel{2U_T}} = \frac{\kappa(V_b - V_{T0})}{\cancel{2U_T}}$$

$$\Rightarrow \kappa V_{in} - V_{out} = \kappa V_b$$

$$\Rightarrow V_{out} = \kappa(V_{in} - V_b)$$

B) Cont'd

We need the bias transistor to be in Saturation.

which is when

$$V_{out} \geq V_{DSsat} = \begin{cases} 4U_T & \text{if } I_6 \ll SI_s \text{ (WI)} \\ K(V_6 - V_{T0}) & \text{if } I_6 \gg SI_s \text{ (SI)} \end{cases}$$

$$V_{out} = K(V_{cm} - V_6) \geq V_{DSsat}$$

$$\Rightarrow \boxed{V_{cm} \geq V_6 + \frac{V_{DSsat}}{K}}$$

C) $\boxed{I_{out} = I_1 - I_2}$

D) We need to make the currents equal between M_3 & M_4 and with I_6. if M_3 or M_4 were less than I_6 current would need to flow through both I_1 and I_2 to compensate for the current draw, which wouldn't be flowing through the rest of the circuit and not sufficient current on the output. Or current would flow up through the current mirror transistors, which would break the circuit. if M_3 or M_4 are more it would be high power consumption for the opamp which is not what we want.

E)